Piecewise-function item repair

My role Item editing and math review

I revised a 27-item piecewise-function set to make graph choices and endpoint inclusion precise. The excerpt shows how the answer options differ at the boundaries.

ORIGINAL WORK FILE EXCERPT

Item 1 of 27

Question update (just the final question):

Question update:

Select the graph that matches the piecewise function.

[Keep the function as shown currently.]

Multiple Choice (display each option as a graph on -10 ≤ x ≤ 10 and -10 ≤ y ≤ 10)

A. f(x) = { x^3 if x < -1 ; -2 if -1 ≤ x < 4 ; sqrt(x) if x ≥ 4 }

B. f(x) = { x^3 if x ≤ -1 ; -2 if -1 < x < 4 ; sqrt(x) if x ≥ 4 }

C. f(x) = { x^3 if x < -1 ; -2 if -1 ≤ x ≤ 4 ; sqrt(x) if x > 4 }

D. f(x) = { x^3 if x < -1 ; 2 if -1 ≤ x < 4 ; sqrt(x) if x ≥ 4 }

Answer: A